UX Primer for Frontend Devs

An abridged field guide to the principles behind good interfaces, with examples from fomodoro. Prepared November 2026.

How to read this. Each idea gets a plain-English definition and an example from our app (in green). You do not need to memorize any of it. The goal is to recognize these patterns when you review a screen or a PR and to have a name for what feels off.

1. Nielsen's 10 usability heuristics

Jakob Nielsen (with Rolf Molich) distilled hundreds of usability problems into ten rules of thumb in 1994. They are still the most widely used checklist for reviewing an interface, mostly because they describe how people think rather than how screens look.

Heuristic	What it means	In fomodoro
1. Visibility of system status	Always show what's happening, promptly.	Big countdown, block name, cycle dots, time in the tab title.
2. Match the real world	Use the user's words and familiar concepts, not system jargon.	"Short break", not "Interval type 2".
3. User control and freedom	Give a clear way out of mistakes: undo, cancel, back.	Pause, Skip, Reset. Reset asks nothing because it's cheap to redo.
4. Consistency and standards	Same words and actions mean the same thing everywhere; follow platform conventions.	Space toggles start/pause, just like media players.
5. Error prevention	Better than good error messages: design so the error can't happen.	Steppers instead of free-typed minutes, so "2500" is impossible.
6. Recognition over recall	Show options and state; don't make people remember them.	"Next: short break, 5 min" so you never count blocks.
7. Flexibility and efficiency	Shortcuts for experts that novices can ignore.	Keyboard shortcuts; the mini timer window.
8. Aesthetic, minimalist design	Every extra element competes with the important ones.	One primary button; settings behind one icon.
9. Help users recover from errors	Plain-language errors that say what happened and what to do.	"Code expired. Send a new one." with the button right there.

10. Help and documentation	Ideally unnecessary; when needed, short and in context.	The iPhone install sheet appears exactly when you need it.

2. The "laws of UX"

These come from psychology research. Jon Yablonski's *Laws of UX* (2020, lawsofux.com) popularized them as a designer's vocabulary. The first two are the ones you'll use weekly.

Hick's law: more choices, slower decisions

Decision time grows (roughly logarithmically) with the number of equally weighted choices. Implication: reduce options at moments that need speed, group or hide the rest. It does *not* mean "fewer features"; it means fewer choices *at once*.

The timer screen has one primary action. All configuration lives in Settings.

Fitts's law: big and close is fast

The time to hit a target depends on its distance and size. Bigger and nearer is faster and less error-prone. Screen edges and corners are effectively "infinite" targets on desktop because the cursor stops there.

Large round Pause button near the thumb zone; Reset and Skip smaller and away from it so they're not tapped by accident.

Law	The idea	Practical use
Jakob's law	Users spend most of their time on *other* apps and expect yours to work the same way.	Don't reinvent settings screens, sign-in, or gestures.
Doherty threshold	Productivity soars when the system responds in under ~400 ms.	Optimistic UI: show the change instantly, sync after.
Miller's law	Working memory is limited (the "7 ± 2" figure is often overstated; modern estimates are closer to 4).	Chunk information. Never use it to justify "max 7 menu items".
Tesler's law	Every system has irreducible complexity; someone has to deal with it, ideally the system, not the user.	We handle iOS push quirks so you just see "Break time".
Goal-gradient effect	People speed up as they approach a goal.	The progress ring and cycle dots make the finish line visible.
Zeigarnik effect	Unfinished tasks stick in memory.	Show incomplete progress (gently) to invite completion.
Peak-end rule	Experiences are judged by their peak and their ending.	Make the "block done" moment feel good: a pleasant chime, not a klaxon.

Aesthetic-usability effect	Attractive designs are perceived as easier to use, and get forgiven for small flaws.	Polish matters, but it can hide real usability problems in testing.
Von Restorff (isolation)	The item that differs is the one remembered.	Make exactly one thing visually distinct: the primary action.
Gestalt: proximity, similarity, common region	Things close together, alike, or inside one boundary are seen as a group.	Spacing is a grouping tool; use it before borders and lines.

3. Don Norman's design principles

From *The Design of Everyday Things* (1988, revised 2013), the foundation most of the above builds on. Norman coined "user experience" as a job title at Apple in the 1990s.

- **Affordances and signifiers.** An affordance is what an object lets you do; a signifier is the visible clue. A flat text label that's secretly a button has the affordance but no signifier.
- **Feedback.** Every action gets an immediate, visible response. Pressing Start instantly changes the button to Pause and starts the ring moving.
- **Mapping.** Controls should relate obviously to what they affect. A stepper's minus on the left and plus on the right maps to "less" and "more".
- **Constraints.** Limit what can be done so the wrong thing is hard to do (pairs with Nielsen #5).
- **Conceptual model.** Users build a mental model of how the system works; the design should make the right one easy to form. "A cycle is 4 focus blocks with breaks, then a long break."

4. Progressive disclosure

Definition: show only what most people need for the current task, and make the rest available on request. It manages complexity by *sequencing* it, not by removing it. Nielsen Norman Group has written about it since 1995; it's one of the most reliable ways to make a capable product feel simple.

Common forms

- **On-demand:** secondary options behind a menu, "Advanced" section, or settings icon. Durations, sounds, 2FA.
- **Staged (step-by-step):** a flow split into screens, each with one job. Sign-in: email first, then the code screen.
- **Contextual (just-in-time):** information that appears at the moment it becomes relevant. "Install the app to get notifications" appears only when you turn notifications on in iPhone Safari.
- **Expandable details:** accordions, "Show more", tooltips for explanations.

How to get it right

- **Sort by frequency, not importance.** What people do most goes on top; rare tasks go one level down.
- **Keep it shallow.** More than two levels of hiding and people get lost. If you need three, restructure.
- **Make the path visible.** The hidden content needs a clear signifier ("More options", a gear icon) or it effectively doesn't exist.
- **Don't hide what people need to finish the task**, especially errors, prices, or consequences. That crosses into deceptive design.

> **Same idea, applied to Claude.** Our context-management rules (short AGENTS.md, rules that load only for matching files, skills that load on demand) are progressive disclosure for an AI. The principle is general: show the essential first, the rest on request.

5. Apple Human Interface Guidelines (HIG)

Apple's official design guidance for its platforms (developer.apple.com/design/human-interface-guidelines). Even for a web app, following it makes the app feel native on iPhone and Mac. Since the 2025 "Liquid Glass" redesign, Apple frames its principles as **hierarchy** (content first, controls recede), **harmony** (UI that fits the hardware's shapes), and **consistency** across devices.

Guideline	Rule of thumb
Touch targets	At least 44 × 44 points. Most common iPhone web-app mistake.
Safe areas	Keep content clear of the notch, Dynamic Island, and home indicator (CSS `env(safe-area-inset-*)`).
Dynamic Type	Text should scale with the user's text-size setting. On the web: size text in `rem`, never fixed px everywhere.
Dark Mode	Support it with real dark palettes, not just inverted colors.
Standard gestures	Don't override system gestures (edge swipe for back, swipe up for home).
Notifications	Concise, actionable, never marketing. Users disable apps that nag.
Motion	Purposeful and brief; honor "Reduce Motion" (`prefers-reduced-motion`).

Google's equivalent is Material Design 3 (m3.material.io). Worth skimming for its component guidance even if we don't use it.

6. WCAG 2.2 (accessibility)

The Web Content Accessibility Guidelines from the W3C. Version 2.2 (October 2023) is current; WCAG 3 is a long-running draft. Levels are A, AA, AAA, and **AA is the standard target** (it's what most laws reference). Everything hangs off four principles, remembered as **POUR**:

- **Perceivable:** content can be seen or heard (text alternatives, contrast, captions).
- **Operable:** everything works by keyboard, there's enough time, nothing triggers seizures.
- **Understandable:** readable text, predictable behavior, help with input errors.
- **Robust:** works with assistive technology, which mostly means correct semantic HTML.

The AA criteria you'll hit most

Criterion	Requirement
Contrast (1.4.3)	4.5:1 for normal text, 3:1 for large text and UI components.
Use of color (1.4.1)	Color can't be the only way information is conveyed. Why "Focus" stays as text on the tinted screen.
Keyboard (2.1.1)	Everything works without a mouse; focus is always visible.
Status messages (4.1.3)	Screen readers announce changes without moving focus (ARIA live regions). "Short break started."
Reflow (1.4.10)	Works at 320 px wide / 400% zoom without horizontal scrolling.

New in 2.2

- **Focus not obscured:** the focused element can't be hidden behind sticky headers or banners.
- **Target size (minimum):** at least 24 × 24 CSS px (Apple's 44 pt is stricter; we follow Apple).
- **Dragging movements:** anything draggable also needs a non-drag alternative.
- **Accessible authentication:** no memory or puzzle tests to sign in; allow paste and password managers. Our 6-digit code field must allow paste and use `autocomplete="one-time-code"` so iOS offers the code from Mail.
- **Redundant entry** and **consistent help:** don't make people retype what they already entered; keep help in the same place.

> **The first rule of ARIA:** use a native HTML element (`<button>`, `<input>`, `<dialog>`) before reaching for ARIA attributes. Native elements come with keyboard and screen-reader behavior built in; a `<div onclick>` does not.

7. Modern frontend practice: the rest of the toolkit

- **Performance is UX.** Google's Core Web Vitals measure it: **LCP** (main content visible, ≤ 2.5 s), **INP** (response to taps and clicks, ≤ 200 ms), **CLS** (layout doesn't jump, ≤ 0.1). Tabular digits on the timer prevent tiny layout shifts every second.
- **Design systems and tokens.** Colors, spacing, and type defined once as named variables (`--color-focus`, `--space-4`) and reused everywhere. Consistency becomes automatic and dark mode becomes a token swap.
- **Mobile-first, responsive.** Design the smallest screen first, then add. Container queries let components adapt to their own space, not just the window.
- **Respect user preferences.** `prefers-color-scheme`, `prefers-reduced-motion`, text size, Focus modes.
- **Visual hierarchy through restraint.** Size, weight, color, and space, in roughly that order. Most "it looks amateur" problems are spacing and too many competing weights. *Refactoring UI* is the best practical book here.
- **Avoid deceptive patterns** (formerly "dark patterns"): fake urgency, hidden costs, guilt-shaming opt-outs, roach-motel cancellation. Increasingly regulated (EU DSA, US FTC).
- **Inclusive design.** Microsoft's framing: "solve for one, extend to many". Captions help deaf users *and* people in loud rooms; big targets help motor impairments *and* people walking.
- **Test with real use.** Five-user usability tests catch most major issues (Nielsen). For a single-user app, the equivalent is dogfooding plus a short note whenever something annoys you.

8. Further reading, in order

- **Steve Krug, *Don't Make Me Think***: the fastest, most readable intro. Start here.
- **Nielsen Norman Group (nngroup.com/articles)**: free, research-backed articles on everything above.
- **lawsofux.com**: one page per law, with sources.
- **Wathan and Schoger, *Refactoring UI***: practical visual design for developers.
- **Don Norman, *The Design of Everyday Things***: the foundational theory.
- **Apple HIG** and **WCAG 2.2 Quick Reference (w3.org/WAI/WCAG22/quickref)**: reference, not cover-to-cover reading.
- **web.dev**: Google's guides on performance, PWAs, and accessibility.